

RailVision EcoRail Strategic Opportunity Brief

RailVision Strategy Brief

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CONFIDENTIAL STRATEGIC ANALYSIS

1. Executive Summary (BLUF)

RailVision Analytics is building EcoRail, an AI-enabled, self-driving technology platform for trains that begins with a high-ROI “Google Maps for rail” product and expands toward semi- and full-autonomy. Using AI and self-driving algorithms, EcoRail delivers real-time driving recommendations, fuel savings, and emissions reductions, positioning it as core operating technology for the rail industry.

Strategically, the near-term opportunity is to land and expand with high-value railroad groups—starting with large multi-railroad owners like G&W (owns 120 railroads) and Watco (owns 45 railroads), and marquee passenger and Class II operators like Via Rail and Wheeling & Lake Erie. These accounts collectively represent tens of millions in ARR potential at current capability levels, and a pathway into the broader EcoRail automation roadmap.

EcoRail's algorithms have demonstrated meaningful fuel savings compared to legacy cruise control systems, and the overall rail industry is undergoing a digital revolution, creating strong tailwinds for adoption of AI-powered operations automation. With limited hardware (tablet + bracket), onboarding in weeks, and locomotives staying in service during deployment, RailVision can scale rapidly across fleets while maintaining strong unit economics.

This brief summarizes: (i) EcoRail's current product and capability stack, (ii) priority customer opportunities and ARR potential, (iii) market size and automation roadmap, (iv) competitive positioning, and (v) recommended next steps to accelerate enterprise sales.

2. RailVision & EcoRail Overview

RailVision Analytics is an AI company focused on rail operations automation. Its foundational product, EcoRail, is an AI-enabled solution that acts as a 'Google Maps' for the rail industry, using AI and self-driving algorithms to make real-time recommendations that save fuel and reduce greenhouse gas (GHG) emissions.

Core context from internal materials:

- RailVision is taking a staged, revenue-generating approach to build the foundational platform of autonomy for railroads.
- The RailVision Analytics product is positioned as core operating technology for railroads, with three product tiers already built.
- EcoRail is already a revenue-generating AI-enabled solution.

EcoRail's core value proposition:

- Fuel and emissions reduction: RailVision reduces a railroad's fuel costs and emissions by up to 20% (supported by solution slides), with demonstrated savings delivery of around 7% in documented programs.
- AI-powered operations: AI and self-driving algorithms, specifically trained for rail industry routing and driving recommendations, provide real-time train-handling guidance.
- Low-friction deployment: EcoRail uses limited hardware (a tablet and bracket) and integrates with existing trains and track systems, enabling rapid deployment.

EcoRail fits into the broader Rail operations automation market, acting as the initial step toward semi-autonomous and fully autonomous operations while generating immediate operational savings for customers.

3. Strategic Customer Opportunity

RailVision's near-term growth can be anchored in a focused set of high-impact customers that control large networks or offer strong signaling value to the market.

3.1 Priority Freight & Holding Company Targets

1) G&W (Genesee & Wyoming)

- Profile: Owns 120 railroads.
- ARR Potential: \$10M identified ARR potential.
- Strategic Rationale:
- High leverage: a single commercial relationship unlocks deployment across a large, diverse portfolio of short line and regional railroads.
- Ideal use case for EcoRail algorithms that can standardize best-practice train handling across many operating environments.
- Platform account for iterative rollout of manual mode, supervised self-driving, and ultimately higher levels of autonomy.

2) Watco

- Profile: Owns 45 railroads.
- ARR Potential: \$5M identified ARR potential.
- Strategic Rationale:
- Mid-sized but diversified network where RailVision can prove rapid scaling and standardized onboarding.
- Strong fit for a limited hardware + software delivery model that minimizes capital outlay and leverages SaaS ARR.

3.2 Passenger & Class II Railroads

3) Via Rail

- Profile: Largest passenger railroad in Canada.
- ARR Potential: \$3.5M identified ARR potential.
- Strategic Rationale:
- High-visibility passenger operator where fuel savings and emissions reduction have both economic and political value.

- Strong reference logo for future passenger market pursuits and for national rail strategies.

4) Wheeling & Lake Erie

- Profile: Major Class II Railroad.
- ARR Potential: \$1.5M identified ARR potential.
- Strategic Rationale:
- Demonstrates EcoRail's fit for non-Class I freight, complementing potential future work with large Class I's like Union Pacific and CN Rail (identified in adjacency context).

3.3 Strategic Implications

- Collectively, these four accounts represent ~\$20M in specific ARR potential under current product scope.
- They provide a cross-section of freight, passenger, and regional networks, ideal for validating EcoRail across different operational contexts.
- As EcoRail progresses from manual to supervised self-driving modes, upsell paths within each customer can increase average monthly revenue per customer via:
- More advanced algorithms and
- Increased network coverage across their fleets.

4. Product & Capability Stack

EcoRail's product design is intentionally staged to deliver immediate value while building toward higher automation.

4.1 Core Workflow: Trip Setup !' Live Recommendations !' Trip Review

- Trip Setup
 - Drivers receive and review all operational information required to operate the train via AI integrations.
 - Integrates timetable, speed restrictions, and operational context into a single, easy-to-understand interface.
- Live Recommendations
 - Drivers receive real-time recommendations for efficient train-handling adapted to all operating conditions.
 - Recommendations are driven by RailVision's self-driving train algorithm, part of its AI-powered self-driving technology for trains.
- Trip Review
 - After each trip, drivers receive a detailed analysis of their operational performance, closing the loop for continuous improvement and training.

4.2 In-Cab Decision Support Interface

The EcoRail interface provides a set of core, high-signal data elements for the crew:

- Driving Recommendation (what to do now)
- Speed Limit (regulatory/track limit)
- Actual Speed (current performance)
- Elevation View (upcoming grades/terrain)
- Train on Map (location context)
- Estimated Time of Arrival
- Temporary Speed Restrictions

This interface functions as a rail-specific navigation and decision-support system—the effective ‘Google Maps for rail’ experience.

4.3 AI & Algorithmic Capabilities

- AI and self-driving algorithms tuned for rail routing and train-handling recommendations.
- Fuel savings algorithms that optimize throttle and braking to reduce fuel burn while meeting schedule and safety constraints.
- EcoRail algorithms delivering up to ~25% energy savings potential vs baseline in modeling, materially outperforming legacy competitor cruise control algorithms (around 5% and developed in the early 1990s).
- Self-driving train algorithm is the core engine for supervised self-driving and future autonomous modes.

4.4 Hardware & Deployment Model

- Limited hardware consisting of a tablet and bracket.
- Trains are retrofitted to operate in existing track systems using AI, avoiding costly infrastructure overhauls.
- Because the footprint is minimal, RailVision can:
- Rapidly scale the solution across fleets.
- Maintain low installation time/cost per locomotive.

Collectively, this stack positions EcoRail as a modern, AI-native operations layer that can sit on top of legacy rail infrastructure with minimal friction.

5. Market Size, Automation Roadmap & Revenue Potential

EcoRail participates in the broader Rail operations automation market and is structured to capture value as railroads progress along an autonomy continuum.

5.1 Market Context & TAM

- The rail industry is described internally as undergoing a digital revolution, creating structural demand for data-driven and AI-based operating technology.
- EcoRail's total ARR potential at full automation is estimated at \$12B–\$19B, reflecting a large global/continental TAM for AI-enabled autonomous train operations.
- Within this, the EcoRail Supervised Self-Driving segment alone is sized at ~\$3B ARR.

5.2 Automation Roadmap & Product Modes

EcoRail is designed with clear capability tiers that match the rail industry's risk and adoption profile:

1) EcoRail Manual Mode

- AI provides recommendations, but humans remain fully in control of operations.
- Easiest to deploy from a regulatory and labor perspective; critical for initial adoption.

2) EcoRail Supervised Self-Driving

- AI takes a more active control role under human supervision.
- Properties:
 - Maximum fuel savings: up to 15% in fuel savings.
 - Additional savings from reduced equipment wear (smoothed driving profiles).
- Segment market size ARR ~ \$3B.

3) Semi-autonomy and Full-autonomy (conceptual higher tiers)

- Internal capabilities reference semi-autonomy and full-autonomy as target states.
- As algorithms and network coverage expand, average monthly revenue per customer increases.

5.3 Revenue Model & Options

- RailVision's AI-enabled solution is revenue-generating today, consistent with SaaS/usage-based pricing anchored in demonstrated fuel savings.
- A Flat-rate option exists in the commercial model, typically paired with mechanisms like:
- Savings delivery guarantees, e.g., ~7% demonstrated savings against a recalibrated annual baseline.
- Annual baseline recalibration to reset expected fuel consumption benchmarks and preserve ongoing savings attribution.

5.4 Strategic Implications

- Near-term revenue growth is driven by manual mode deployments across key freight and passenger networks (e.g., G&W, Watco, Via Rail, Wheeling & Lake Erie).
- Medium term, upsell into supervised self-driving across already-landed accounts substantially expands ARR per locomotive and per customer.
- Long term, RailVision is positioned to capture a portion of the \$12B–\$19B EcoRail automation TAM as the industry transitions from decision-support to autonomy.

6. Competitive Positioning & Adjacent Benchmarks

EcoRail competes against legacy rail technologies and is comparable to leading autonomy players in adjacent transportation sectors.

6.1 Legacy Rail Competitors

- The primary legacy solution class is cruise control algorithms, developed in the early 1990s.
- These systems deliver ~5% energy savings under ideal conditions and are not deeply integrated with modern AI or real-time data.

EcoRail's advantages vs cruise control:

- Smarter algorithms (EcoRail algorithms) capable of up to ~25% energy savings potential in modeling versus baseline, and documented 7% savings delivery in live deployments.
- Designed to integrate with existing track systems and real-world operational data, rather than static control logic.
- Simple, enjoyable user experience that is liked by crews, increasing adoption and adherence.

6.2 Digital & Autonomy Peers (Adjacent Benchmarks)

Internal context references adjacent companies:

- Nexus Robotics, Embark Trucks, and Applied Intuition—all associated with high-autonomy robotics, autonomous trucking, and simulation/validation.
- Union Pacific Railroad and CN Rail appear in the context as large, innovation-focused railroads that signal where the market is heading.

Positioning takeaways:

- RailVision is to rail what autonomous trucking platforms are to road freight—an AI-native operations layer and automation stack.
- EcoRail's staged, revenue-generating approach mirrors strategies used by high-growth autonomy startups: start with decision-support, then move to supervised autonomy.

6.3 Strategic Differentiators

- AI-first architecture focused specifically on rail operations, not retrofitted from generic optimization tools.

- Rapid deployment, limited hardware, and no need for fundamental infrastructure change.
- Strong alignment with customer priorities: fuel savings, emissions reduction, and operational consistency across fleets.

This combination positions RailVision as a category-defining player in rail operations automation rather than a feature-level competitor to legacy systems.

7. Implementation, Onboarding & Scaling Model

A key strategic advantage for RailVision is the ability to implement and scale EcoRail with minimal disruption and capital expenditure.

7.1 Onboarding Process

- Onboarding in weeks: EcoRail deployments are designed to be completed in weeks, not months, accelerating time-to-value.
- Locomotives stay in service during onboarding:
- Installations and configuration can be performed without sidelining locomotives for extended periods.
- Reduces opportunity cost and makes pilots and rollouts easier to sell internally for railroads.

7.2 Hardware & Integration

- Limited hardware: a tablet + bracket form factor.
- Integrates via AI integrations that pull in the operational information required to operate the train (timetables, restrictions, etc.).
- Retrofitting trains to operate in existing track systems ensures compatibility with current infrastructure and avoids large capex.

7.3 User Experience & Change Management

- The system is designed to provide a simple, enjoyable user experience that is liked by crews.
- Crews and supervisors use a dashboard to monitor their savings and performance, reinforcing behavior change with clear feedback.

7.4 Scaling Across Networks

- Once a customer is onboarded, RailVision can rapidly scale the solution:
- Expand from initial pilot fleets to network-wide deployment.
- Increase network coverage and attach more advanced algorithms over time.

This implementation model underpins a scalable go-to-market motion: land with quick-deploy decision-support, then expand coverage and autonomy levels within each railroad.

8. Recommended Next Steps for Enterprise Sales

To fully exploit the EcoRail opportunity within the current strategic and market context, the following actions are recommended:

8.1 Prioritize High-Leverage Accounts

- Sequence focus:
 - 1) G&W (120 railroads, \$10M ARR potential).
 - 2) Watco (45 railroads, \$5M ARR potential).
 - 3) Via Rail (largest Canadian passenger railroad, \$3.5M ARR potential).
 - 4) Wheeling & Lake Erie (Major Class II, \$1.5M ARR potential).
- Position these as flagship case studies representing freight, passenger, and regional markets.

8.2 Land with Manual Mode, Design Path to Supervised Self-Driving

- Commercially lead with EcoRail Manual Mode:
- Emphasize quick deployment, no service downtime, and fuel/emissions savings.
- Architect contracts and pilots to include clear milestones for:
 - Upgrading to EcoRail Supervised Self-Driving.
 - Expanding network coverage across more locomotives and lines.

8.3 Align Pricing with Savings Delivery

- Use a combination of:
 - Flat-rate options for simplicity, and
 - Savings delivery metrics (e.g., targeting ~7% demonstrated savings with an annual baseline recalibration) to support ROI narratives.
- For large holding companies, explore portfolio-wide pricing structures that reward rapid scale-up.

8.4 Strengthen Competitive Narrative

- Clearly contrast EcoRail against legacy cruise control algorithms:
- Highlight the 5% vs 25%+ energy savings potential gap.

- Emphasize AI-native approach and modern UX that crews prefer.
- Use adjacent autonomy leaders (e.g., Embark Trucks, Applied Intuition) as benchmarks when communicating with investors and innovation-focused railroads (UP, CN).

8.5 Build a Reference Flywheel

- Secure at least one marquee case study in each of the following:
- A large multi-railroad holding company (e.g., G&W or Watco).
- A national passenger operator (Via Rail).
- A prominent regional/Class II freight railroad (Wheeling & Lake Erie).
- Measure and publish:
- Fuel and emissions reductions.
- Operational performance metrics (on-time performance, smoother driving profiles).

Executing on these steps will enable RailVision to convert its strong product and market position into durable, high-ARR relationships and establish EcoRail as the default AI operations layer for railroads pursuing digital and autonomy strategies.

Introduction (Updated BLUF)

This brief provides a CEO-ready decision frame for evaluating Holland as a non-exclusive co-sell partner to accelerate RailVision's penetration of North America shortlines and regionals, while preserving our direct customer relationship, data rights, and long-term strategic options.

Bottom Line Up Front

- Decision in first 90 days: Go / No-Go on scaling a Holland co-sell motion based on concrete results, not intent.
- Targets: \1\ 3 signed POCs and \2\ 1 production contract within 90 days, with statistically significant fuel and/or time savings and high crew acceptance.
- Guardrails (non-negotiables):
 - Non-exclusive, co-sell only: Holland is a prospective co-sell partner; no exclusivity, no reseller role, no white-label.
 - RailVision controls economics & relationship: We hold SOWs, set price, and own renewal; Holland cannot invoice or set pricing.
 - Data rights preserved: RailVision must receive trip, usage, and baseline performance data for all joint deployments.
 - Positioning discipline: RailVision remains an “engineer-first efficiency companion”—explicitly not autopilot, not surveillance, and not a tool to replace engineers.

Why Holland Matters

- Holland already provides maintenance and services and has deep relationships with shortlines, giving them trusted field access where we are still building presence.
- They can layer RailVision's tablet-first, low-capex, rapid deployment capability onto existing site visits, enabling faster pilots without locomotive surgery.
- Their field services (install assistance, crew coaching, change-management) directly support our need for high crew adoption and behavior change on the ground.

What Success Looks Like

- Commercial:
 - "e 3 POCs signed and "e 1 production contract within 90 days.
 - Growing qualified pipeline ("e5 qualified deals) across shortlines/r
- Adoption & Operations:

- Crew adoption "e 70% active use in POCs; opt-out rate "d 10%.
- Time-to-first-train "d 14 days from SOW signature.
- Proof & Story:
 - Documented fuel and time savings with confidence intervals.
 - At least 2 anonymized case snapshots for future sales enablement and fundraising.

The sections that follow detail: (1) the proposed co-sell model and compensation structures, (2) the 90-day joint GTM and enablement plan, (3) metrics and scorecard for the Holland partnership, and (4) risks, constraints, and the explicit Go / No-Go bar for scaling this channel.